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Adult–Adult Play in Primates: Comparative Analyses of Its Origin, Distribution and Evolution

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Abstract

Comparative analyses were conducted on a data set derived from the literature so as to test several hypotheses which were developed to explain the distribution of adult–adult play fighting within the order primates. Ratings for play occurring in sexual and non-sexual contexts were developed. Three hypotheses were evaluated: (i) that play occurring in non-sexual social contexts is a byproduct of its use in sex; (ii) that the occurrence of play is related to its use for social assessment and manipulation, and so is more likely to be present in species with reduced familiarity between individuals; and (iii) that phylogenetic affiliation influences the likelihood that species within clades engage in play. We used independent contrasts to test the first two hypotheses, and both were significant, with the presence of play in sexual contexts accounting for 14–16% of the variance of play in non-sexual contexts, and reduced social familiarity accounting for 30–40% of the variance in the occurrence of play in non-sexual contexts. To test the third hypothesis, we mapped the occurrence of both types of play onto known phylogenies. The overlap was not congruent, indicating that phylogenetic relationships did not account for the distribution of play. Given that play in both sexual and non-sexual contexts was more likely to occur in species with a social organization involving reduced frequency of contact between the sexes and other social group members, we suggest that the likely adaptive value of play fighting is as a tool for social assessment and manipulation. The possible factors that mitigate the use of play fighting for these purposes, such as the availability of other forms of communication that could serve similar functions, are discussed.

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Introduction

Play is most commonly reported in juveniles (Fagen 1981). Developmental studies consistently show that while play begins in infancy, it reaches its highest frequency in the juvenile phase (e.g. Thor & Holloway 1984). Although the incidence of play is reduced following puberty, in many species it can continue to occur well into adulthood (e.g. Rhine 1973; Epps 1974; Pellis & Pellis 1991). Given that most play occurs in the juvenile phase, efforts to study and explain play have been primarily focused on childhood (e.g. Smith 1984; Bekoff & Byers 1998; Power 2000). However, a complete understanding of play requires an account of its occurrence in adulthood (Brueggeman 1978; Sutton-Smith 1997; Burghardt 1998).

The study of play, especially when it is focused on non-human animals, is typically categorized into three general types: locomotor play, object play and social play (Burghardt 1998; Power 2000). But not all species that play, engage in all three categories (Fagen 1981). For example, object play is rare in rodents (Hole & Einon 1984), but common in carnivores (Fagen 1981). Of these categories, the most pervasive form of play is social play (Pellis & Pellis 1998b), and is the form most likely to continue into adulthood (e.g. Rhine 1973; Pellis & Pellis 1991, 1997; Winn 1994). Furthermore, the most commonly reported form of social play involves some kind of competitive interaction referred to as 'play fighting' (Pellis & Pellis 1998b). When retained in adulthood, play most often involves some variation on the juvenile pattern of play fighting (Pellis & Pellis 1998a, b; Pellis & Iwaniuk 1999b).

Play fighting in adulthood most often involves an immature partner (e.g. Brueggeman 1978; Pellis & Pellis 1997); and while sometimes initiated by adults, it is most often initiated by the juveniles (Fagen 1981). In part, adult–juvenile play appears to be influenced by demographic factors, such as a paucity of peers (e.g. Poirier 1970). Even so, species differences in the propensity of adults to play with juveniles do exist (Caine & Mitchell 1979). Because the opportunity to compare a wide range of species under similar conditions is rare, reports of adult–juvenile play are unfortunately difficult to evaluate. Adult–adult play, while less frequent, provides a less ambiguous means of assessing the propensity for the adults of different species to engage in play fighting (Pellis & Iwaniuk 1999b). In the current study, we used comparative methods (Brooks & McLennan 1991; Harvey & Pagel 1991) to test three hypotheses to account for the variance in the distribution of adult–adult play fighting in primates.

One explanation for the occurrence of adult–adult play fighting is that it is a juvenile feature, retained in adulthood for courtship purposes. Courtship play is hypothesized as providing a means of familiarizing potential pairmates. A prediction arising from this hypothesis is that playful contact preceding copulation should occur in species where males and females are solitary, and so unlikely to be familiar with one another (Aldis 1975; Fagen 1981). A comparative analysis of adult–adult play during courtship in primates confirmed this hypothesis. The degree of the species-typical male–female association preceding mating accounted

for 40% of the variance in the distribution of play fighting (Pellis & Iwaniuk 1999b). These findings support the hypothesis that the value of playful contact during courtship has led to the retention of a juvenile pattern into adulthood.

However, in reviewing the primate literature, it was clear that many instances of play fighting among adults were reported for non-sexual contexts (Pellis & Iwaniuk 1999b). One possibility is that if play had been retained into adulthood because of its function in courtship, then playful modes of interaction could have been co-opted for use in non-sexual contexts. If this hypothesis is correct, then the occurrence of play in courtship should significantly predict the occurrence of non-sexual play. It should be remembered, however, that 60% of the variance in the distribution of courtship play among adult primates was not predicted by the pattern of male–female association (Pellis & Iwaniuk 1999b). Therefore, even if the byproduct hypothesis is confirmed, it is likely that a large portion of the variance will remain unexplained. For this reason, two additional hypotheses are proposed that may contribute to explaining the variance in the distribution of adult–adult play.

The first of these subsidiary hypotheses is the 'social assessment hypothesis'. Given that play appears to be useful in the pre-copulatory encounters of pairmates that are unfamiliar with one another (Pellis & Iwaniuk 1999b), it is possible that this function has been generalized as a tactile means of communication and conspecific assessment in a wider range of social contexts in which the interactants' relationship is in a state of flux (Brueggeman 1978). Indeed, non-sexual use of play fighting by adults has been reported for many species of mammals (Pellis & Pellis 1996). The manner and type of information extracted from a partner by play fighting, and the manner and type of manipulation possible with such interactions, appears to vary with context and species. For example, within established relationships, play fighting may be used to maintain affiliative bonds (e.g. Ehrlich & Musicant 1975; Geist 1981; Pellis & Pellis 1991) or to test for weakness and so gain status in the dominance hierarchy (e.g. Geist 1982; Jones 1983; Pellis et al. 1993; Paquette 1994). Alternatively, when encountering unfamiliar animals, play fighting may be used to assess the partner and to establish a dominance relationship without engaging in overt fighting (e.g. Newell 1971; Smith et al. 1999). The kind of information gained from such play, and the circumstances in which play rather than some other strategy is used, is at present unknown. It is also not clear whether such playful interactions are specific to particular functions, or are themselves a subset of a more general class of manipulative social behavior, such as is reflected in the 'testing the bond' hypothesis proposed by Zahavi & Zahavi (1997).

Regardless of the mechanisms and the functional rules involved, if the social assessment hypothesis is true, then it would be predicted that play fighting should be most likely to occur in situations of social uncertainty. One source of social uncertainty may arise from relatively infrequent contact between members of a social group. Therefore, it is predicted that non-sexual play fighting should be more prevalent in species with less cohesive social systems, and so with a less frequent opportunities for social contact.

The second subsidiary hypothesis is the 'phylogenetic inertia hypothesis'. That is, play is plesiomorphic within primates, and has diversified little past its point of origin. Indeed, the nocturnal prosimians have a suite of morphological traits and socio-ecological patterns most like ancestral primates (Martin 1995), and is the group for which adult–adult play seems to be most widespread (Loizos 1967; Jolly 1985; Bearder 1999). Given that the precise meaning of 'phylogenetic inertia', as it pertains to evolutionary processes, is unclear (Schwenk 1995), we use this term here simply to mean that despite diversification within the order, membership in particular clades is a major predictor of the presence of play. If this hypothesis is correct, then phylogenetic relationship should account for some of the variance in the distribution of play fighting.

Methods

Categories of Play and Scoring Methods

Based on the available literature, data on the prevalence of adult–adult play were obtained for 71 species of primates, including representatives from all extant families (Table 1). Adult–adult play occurs both in sexual and non-sexual contexts, and each was evaluated separately. For convenience, these will be termed sex play and non-sex play, respectively. Play in both contexts was categorized on an ordinal scale: rare or absent, 0; occurs on some occasions, 1; is a regular and common feature of the behavioral repertoire, 2. Some difficulties arose when trying to distinguish between sex-related and non-sex-related play in some species. For example, adult–adult play in *Mirza* and *Perodicticus* has only been described in the context of male–female pairs, and is clearly associated with successful copulation (Epps 1974; Pagès 1978). However, for these species, such play continues to occur outside of the mating period (Anderson 1970; Hladik & Charles-Dominique 1974; Pagès 1983). This suggests a more distant linkage to sexual behavior. Therefore, for such species, both the sexual and non-sexual categories of play were scored positively. To reduce the effects of this uncertainty, we also developed an overall play score for each species.

The overall rating incorporated play fighting from both contexts. Play for each species was scored as rare or absent (0), or relatively common (1). That is, any species scoring a 1 or a 2 for either sex play or non-sex play was given a 1. By making the scoring as simple as possible, the overall measure of play prevalence reduced the chances of misinterpreting published reports, and permitted us to include species for which separate evaluations of sex play and non-sex play were not possible. Also, it circumvented the assumption that play in sexual and non-sexual contexts are fundamentally different types of play. Afterall, it may be the case that play fighting is a general purpose behavior that can be used in many different contexts; some sexual and some non-sexual (Brueggeman 1978). Therefore, the overall score provided a simpler and less theoretically derived data set to further evaluate the phylogenetic inertia hypothesis.

Reports for some species categorically state that adult–adult play does not occur (e.g. Kinzey et al. 1977); others report it as either rare or common. For

Table 1: Information on play and social organization for the species used

Species*	Social organization	Type of play context		Play absent (0) or present (1)
		sexual	non-sexual**	
Alouatta caraya	–	0	2	1
Aotus trivirgatus	–	0	1	1
Ateles geoffroyi	–	1	2	0
Avahi laniger	3	0	0	0
Avahi occidentalis	3	0	0	0
Brachyteles arachnoides	–	0	0	1
Cacajao calvus	–	–	2	0
Callicebus moloch	–	0	0	0
Callicebus torquatus	–	0	0	0
Callimico goeldii	–	0	0	1
Callithrix jacchus	–	1	1	0
Cebuella pygmaea	–	0	0	1
Cebus apella	–	0	1	1
Cebus capucinus	–	0	1	0
Cebus olivaceus	–	0	0	0
Cercocebus albigena	–	0	0	0
Cercopithecus aethiops	–	0	0	0
Cercopithecus diana	–	0	0	0
Cercopithecus neglectus	–	0	0	0
Colobus badius	–	0	0	1
Colobus guereza	–	0	1	1
Daubentonia madagascariensis	0	1	0	0
Erythrocebus patas	–	0	0	1
Eulemur fulvus	2	0	1	0
Eulemur mongoz	3	0	0	1
Galago moholi	1	0	2	1
Galago senegalensis	1	0	2	1
Galagoides demidovii	1	1	–	1
Gorilla gorilla	–	0	1	1
Homo sapiens	–	1	2	1
Hylobates lar	–	0	1	1
Hylobates syndactylus	–	0	1	1
Lagothrix lagotricha	–	0	2	1
Lemur catta	2	0	1	0
Leontopithecus rosalia	–	0	0	1
Lepilemur edwardsii	0	2	2/3	1
Lepilemur mustelinus	0	2	2/3	1
Loris tardigradus	0	2	2/3	1
Macaca arctoides	–	0	2	0
Macaca fuscata	–	0	0	0
Macaca mulatta	–	0	0	0
Macaca nigra	–	0	0	1
Macaca radiata	–	0	1	0
Macaca tonkeana	–	0	0	0
Mandrillus sphinx	–	0	0	0
Microcebus murinus	1	0	0	1
Mirza coquereli	0	2	2	1
Nycticebus coucang	0	2	2/3	1

Table 1: continued

Species*	Social organization	Type of play context		Play absent (0) or present (1)
		sexual	non-sexual**	
Otolemur crassicaudatus	1	0	2/3	1
Pan paniscus	–	1	2	1
Pan troglodytes	–	1	2	0
Papio anubis	–	0	0	0
Papio cynocephalus	–	0	0	0
Papio hamadrayas	–	0	0	0
Papio ursinus	–	0	0	1
Perodictus potto	0	2	2/3	1
Phaner furcifer	0	–	1	1
Pithecia pithecia	–	0	1	1
Pongo pygmaeus	–	1	0	0
Presbytis entellus	–	0	0	0
Presbytis johnii	–	0	0	1
Propithecus verreauxi	2	0	1	1
Pygathrix nemaeus	–	1	1	1
Saguinus fuscicolis	–	0	1	0
Saguinus mystax	–	0	0	1
Saimiri sciureus	–	1	1	0
Tarsius bancanus	–	0	0	1
Tarsius spectrum	–	0	1	1
Tarsius syrichta	–	1	0	0
Theropithecus gelada	–	0	0	1
Varecia variegata	3	0	1	

*References used for most species can be found in Fagen (1981), Pellis & Iwaniuk (1999b, 2000), and Power (2000). Additional reference material for other species was derived from Wharton (1950), Hill et al. (1951), Simonds (1965), Chalmers & Rowell (1971), MacKinnon (1974), Oates (1977), Box & Morris (1980), MacKinnon & MacKinnon (1980), Oswald & Lockard (1980), Mellen et al. (1981), Byrne et al. (1983), Clark (1985), Koyama (1985), Eibl-Eibesfelt (1989), Thierry et al. (1989), Robbins (1996), Perry (1996), (1997), (1998), Kinzey (1997), Reichard & Sommer (1997), Verbeek & de Waal (1997), and other references cited in the text.

**For species with two values shown, the first was used to test the byproduct hypothesis and the second to test the social assessment hypothesis (see text).

example, in sexual contexts, play is reported as preceding only some copulations in Demidoff's bushbaby, *Galagoides demidovii* (Charles-Dominique 1977) and the orangutan, *Pongo pygmaeus* (MacKinnon 1974), but is ubiquitous in the courtship in Coquerel's mouse lemur, *Mirza coquereli* (Pagès 1978) and the potto, *Perodicticus potto* (Epps 1974). In some cases, several reports needed to be contrasted to ascertain the relative frequency of play. For example, adult–adult play in rhesus macaques, *Macaca mulatta* is rare (e.g. Symons 1978), requiring protacted study to record these interactions (Brueggeman 1978). In contrast, even a relatively short

study reveals that such play occurs at a high frequency in the stump-tailed macaque, *M. arctoides* (Rhine 1973). Wherever possible, multiple reports on the same species were used to verify the species' scores. Furthermore, our own observations, based on samples of 20–50 h each, on captive colonies of several species, confirmed the validity of the ratings: *Erythrocebus patas*, *Ateles geoffroyi*, *Lemur catta* (Pellis & Pellis 1997), *Hylobates lar*, *Gorilla gorilla*, *Macaca fuscata*, *M. tonkeana*, *Mandrillus sphinx*, *Propithecus verreauxi* and *Varecia variegata* (Pellis & Pellis, unpubl. data).

In interpreting the results of this study, it is important to bear in mind the limitations of the rating scales adopted: they only incorporate the prevalence of play fighting, not the complexity or content of the interactions themselves (Pellis & Iwaniuk 1999b). Indeed, types of play fighting may differ in content. For example, in some ground squirrels, play fighting may involve sequences of sexual behavior or sequences of agonistic behavior (Nunes et al. 1999). In such situations, there is a descriptive basis for labeling some playful encounters as sexual and some as aggressive. However, in the currently available literature on primates, little information is available on the content of play fighting. Furthermore, the wider the range of species compared, the more difficult it becomes to contrast comparable elements of behavioral performance from disparate reports by different authors. Therefore, given the data available in the current literature, comparisons based on the content of play, although desirable, are not feasible.

In order to test the social manipulation hypothesis, a measure of social organization was needed. In particular, the social organization score had to be sensitive to the relative degree of familiarity among group members. Social organization in primates ranges from relatively solitary individuals to large permanent groupings of many males and many females, with the vast majority of simians having relatively long lasting social relationships that involve daily contact (Jolly 1985). For some simians, such as chimpanzees (*Pan* spp.) and spider monkeys (*Ateles* spp.), troops may form a cohesive unit. However, different combinations of individuals may break up into subgroups for foraging and resting purposes, so that all troop members may only meet occasionally (i.e. fission–fusion). Therefore, it is difficult to assign objective scores that capture the relative degree of social familiarity among the various forms of permanent and semi-permanent social groupings. The strepsirrhines provided an easier taxanomic grouping within which to assign a social organization score that served the purposes of testing the second hypothesis.

The strepsirrhines have two major forms of social grouping: gregarious and non-gregarious (Charles-Dominique 1995). In the non-gregarious pattern, the adults have their own individual, but overlapping territories. They forage individually during the night, but then, to varying degrees, associate in common sleeping areas (Bearder 1986). This 'noyau' pattern is found among various nocturnal strepsirrhines. Most diurnal strepsirrhines form gregarious groupings which continue to associate both during feeding and resting. However, between these two extremes there are intermediate states in the degree of gregariousness (Martin 1995).

For the noyau species, two levels of gregariousness are evident. In the first, the animals maintain individual territories, and only exhibit tolerance for meetings with the opposite sex at locations of territorial overlap (Charles-Dominique & Bearder 1979). Furthermore, resting-period contact tends to be sporadic, rather than nightly. This category of social organization exhibits the least degree of gregariousness and so was given a score of 0. In the second noyau category, animals meet more regularly at common overlapping locations during both rest periods and when foraging. Areas of territorial overlap can involve multiple adult females, usually forming a matriarchy. Not only do combinations of this matriarchy spend time together during daily nesting, the nesting group may also include an unrelated dominant adult male and, in some cases, even additional, subordinate males (Charles-Dominique 1977). This category has a more gregarious pattern of association than the first, and so was given a score of 1.

The truly gregarious species have social systems which involve the group members continuously associating in all phases of the daily cycle, but here there are also at least two categories of gregariousness. The first includes those species that form multimale/multifemale groupings. While these species generally remain in the same groups, they may also have inter-group social relationships, especially during the breeding season (e.g. Jolly 1966; Richard 1992). This category was assigned a score of 2. The second category includes species forming a more restricted family grouping, involving the breeding pair and their offspring, and was assigned a score of 3.

It should be noted that these scores are not absolute, but rather should be viewed as being relative to the variation present in the species being compared. For example, *Mirza* may have a degree of 'pre-gregariousness' (Pagès 1978) that is closer to the pattern present in *Microcebus* than *Perodicticus*. The same is true for *Nycticebus*, which may be more gregarious than *Perodicticus* and *Loris* (Roonwal & Mohnot 1977). Nonetheless, on a relative ranking score, *Loris*, *Mirza* and *Nycticebus* are more similar to *Perodicticus* than to *Microcebus* and *Galago*. Clearly, a more accurate measurement system would place these species along a continuous gradient. However, at present, there are insufficient data on most species to allow for a more refined scoring system. *Avahi* also needs special mention because, even though they are nocturnal with family members feeding apart, they form highly cohesive, tight family associations during rest periods (Warren & Crompton 1997). Pair-mates also appear to use auditory signals to maintain contact with one another during foraging (Charles-Dominique 1978). Therefore, in many regards, these species exhibit a degree of group cohesiveness more typical of the diurnal monogamous lemurids. Again, given the objective of this study, where the relative degree of association and familiarity is the issue, placement of these species in the most gregarious category seemed appropriate.

In order to test the social assessment hypothesis using the strepsirrhines, a modification to the non-sex play rating scale was used. In order to compare across the whole suite of primate species, the uppermost rating was a 2, where play is judged to be a regular and common feature of the species repertoire. This sets an artificial ceiling to the scale, which was necessary in order to combine disparate

reports on widely diverse species. For example, *Ateles* and *Pan* were the most playful anthropoids, but it was not possible to judge whether their play could be assessed as more or less prevalent than some of the prosimians. However, in the more restricted data set of the strepsirrhines used to test the social assessment hypothesis, further distinctions could be made among the species with the highest play prevalence scores. That is, compared to the *Galago* spp. and *Otolemur*, which scored a 2, some of the other nocturnal species were clearly more playful. Therefore, for this comparison, the scores for these species were adjusted by increasing the ceiling value so as to better reflect these differences (Table 1).

Comparative Methods

We chose two methods for use in this study, based upon the robustness of the tests and their suitability for testing the proposed hypotheses (Harvey & Pagel 1991; Martins & Garland 1991). We used the Felsenstein (1985) method of independent contrasts to test both the byproduct and social assessment hypotheses. Briefly, this method assesses the strength of correlated evolution between two traits by comparing values at each node throughout a phylogeny, and standardizing them against a measure of variance derived from the branch lengths of the phylogeny (Harvey & Pagel 1991). A speciational (i.e. equal branch lengths) model of evolutionary change was chosen because of a lack of information on the lengths of the branches in the phylogeny. Since the use of estimated branch lengths can reduce or bias the results of comparative analyses (Martins & Hansen 1996), we assumed that all branch lengths were equal.

We used the software package CAIC (Purvis & Rambaut 1995) to calculate the contrasts. The contrasts were checked for proper standardization by regression of their absolute values against the square-root of the sum of the branch lengths (Garland et al. 1993). We then tested the contrasts for normality using the Shapiro–Wilk W-test (JMPIN vs. 3.2.1, SAS Institute). Finally, we regressed the contrasts against one another through the origin (Garland et al. 1992).

To test the phylogenetic inertia hypothesis, we mapped the play scores onto the phylogenies using the parsimony algorithm of MacClade (v.3.05) (Maddison & Maddison 1992). The multi-state scores for sexual and non-sexual play were treated as both ordered and unordered characters, whereas the binary score of the presence/absence of play was treated as unordered only. Polytomies, where present, were treated as unresolved parts of the phylogeny as opposed to radiative speciation events (Maddison & Maddison 1992). The consistency index, which is a measurement of the degree of homoplasy (Maddison & Maddison 1992), was recorded for all of the play scores and compared against the expected consistency index for the number of taxa included (Sanderson & Donoghue 1989). Consistency index values close to one suggest that the distribution of the trait closely matches the phylogenetic relationships in the tree, whereas scores close to zero indicate that relatedness has little bearing on the trait's distribution. Finally, the mapping of traits using McClade also permitted the reconstruction of the most plausible ancestral states for each of the play scores.

In order to illustrate the probable phylogenetic history of play, genus-level cladograms for sex play and non-sex play are shown in the Results section. To resolve variable character states within genera, smaller trees for those genera were used to map the distribution of play, and so determine the plesiomorphic state for that clade. These data were then incorporated into the larger trees. It should be noted, however, that the statistical evaluation of the consistency index values was based on trees derived from species, not genera.

We used the phylogenetic tree presented by Purvis (1995), which is based on a composite of molecular and morphological data, for the comparative analyses requiring information on species' relatedness (i.e. independent contrasts, MacClade). Several more recent phylogenetic hypotheses have modified the position of several species (Shoshani et al. 1996; Goodman et al. 1998; Ross et al. 1998; Yoder & Irwin 1999). Therefore, to confirm the findings based on the Purvis tree, we repeated the analyses on a modified tree that incorporated the more recent data. For the sake of economy, and consistency with previous publications (Pellis & Iwaniuk 1999b, 2000), only figures using the Purvis tree are presented.

Results

Sex Play and Non-Sex Play: The Byproduct Hypothesis

A significant positive correlation (slope = 0.46) was found between sex play and non-sex play ($F_{1,63} = 10.59$, $r^2 = 0.144$, $p = 0.002$). That is, the incidence of non-sex play increases with the incidence of sex play (Fig. 1a). Given that the distributions of both sex play (Shapiro–Wilk $W = 0.69$, $p < 0.0001$) and non-sex play ($W = 0.91$, $p < 0.0001$) were not normally distributed, and multiple transformations were not successful in normalizing their distributions, a Spearman–rank correlation was also performed. The Spearman–rank correlation corroborated the findings from the least-squares linear regression ($Z = 2.95$, $r_s = 0.372$, $p = 0.003$).

Using the alternate phylogeny, a significant, positive relationship (slope = 0.48) was also found ($F_{1,63} = 12.23$, $r^2 = 0.0.163$, $p = 0.006$). This was further corroborated by a Spearman–rank correlation ($Z = 3.74$, $r_s = 0.471$, $p = 0.0002$).

Sociality and Play: The Social Assessment Hypothesis

Using the phylogenetic tree provided by Purvis (1995), the correlation between social organization and non-sex play in the strepsirrhines tended towards a negative correlation, but this was not significant ($F_{1,16} = 4.00$, $p = 0.063$). The relationship was significant with the removal of the outlier value for *Daubentonia* (slope = -0.898; $F_{1,15} = 10.13$, $r^2 = 0.403$, $p = 0.0062$). Given that the distributions of both social organization ($W = 0.88$, $p = 0.043$) and non-sexual play ($W = 0.79$, $p = 0.002$) were not normally distributed, and multiple transformations were not successful in normalizing their distributions, Spearman–rank correlations were also performed. Whether the data set contained *Daubentonia* or not, no significant correlations were found ($p > 0.05$).

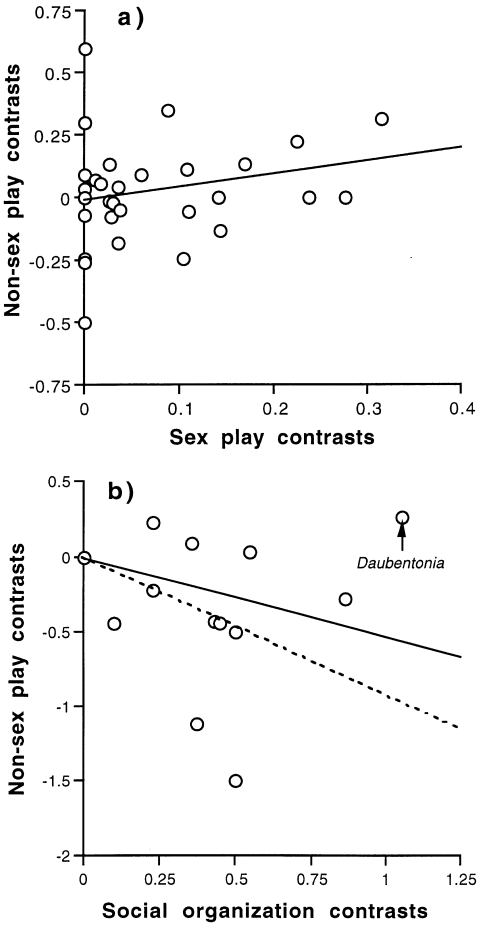


Fig. 1: Scattergrams of the independent contrast comparisons between non-sexual play and (a) sex play (the byproduct hypothesis), or (b), social organization (the social assessment hypothesis). In (b), the solid line represents the regression with *Daubentonia* included; the dashed line represent the regression excluding *Daubentonia*

Using the alternate phylogeny, a significant, negative relationship (slope = -0.696) was also found ($F_{1,16} = 8.16$, $r^2 = 0.324$, $p = 0.011$) even with the presence of *Daubentonia*. This was further corroborated by a Spearman–rank correlation ($Z = 2.14$, $r_s = -0.464$, $p = 0.03$). Overall, the analyses provide some support for the prediction that with increasing social cohesion, the prevalence of play declines (Fig. 1b).

Taxonomic Distribution of Play: The Phylogenetic Inertia Hypothesis

Whether treated as an ordered or unordered trait, the mapping of the play scores yielded low consistency indices using both the Purvis tree (ordered/unordered: sex play = 0.13/0.17; non-sex play = 0.09/0.09; presence or absence of play = –/0.08), and the composite alternative phylogeny (sex play = 0.13/0.17; non-sex play = 0.09/0.10; presence/absence of play = –/0.08). These were all considerably less than the expected consistency index of 0.35 (calculated using a maximum number of 60 species, as per Sanderson & Donoghue 1989). This suggests a relatively large amount of homoplasy and therefore a relatively limited effect of phylogenetic relatedness on the distribution of play. The phylogenetic distribution for both sex and non-sex play are shown in Fig. 2. Note that in this figure, genera are mapped representing 47 terminal branches, not the 60 that were used for statistical evaluation above.

The basal nodes of the trees show that the most likely ancestral state for sex play was for this trait to be rare or absent, whereas, for non-sex play, it was moderately frequent. The alternative composite phylogenetic hypothesis yielded comparable results. Similarly, for the play present/absent score, the reconstruction (not shown) revealed that some play was present in the ancestral node. These patterns suggest that play was likely to be relatively rare to moderate in prevalence in ancestral primates, but was then subjected to further adaptive radiation, with some lineages increasing and others decreasing its prevalence.

Discussion

Our data show that 14–16% of the variance in adult–adult play in non-sexual contexts is significantly predicted by the presence of play in sexual contexts, and so provides support for the hypothesis that adult–adult play is a byproduct of the occurrence of play in sexual contexts as formulated in the Introduction (Aldis 1975; Fagen 1981). However, evaluation of the evolutionary sequence depicted in Fig. 2 suggests the opposite. Non-sexual play appears before sexual play in nearly every case. The presence of sexual play in the absence of non-sexual play in *Daubentonia* and *Pongo* is most likely a reflection of the solitary life-style of these species, which allows for little opportunity for non-sexual play to occur (see below). Furthermore, while sexual play appears to be rare or absent in ancestral primates, non-sexual play appears to be present at a moderate level. These findings suggest that the original function of play fighting among adults involved some form of social assessment and manipulation, and then later co-opted into a courtship in some species.

That a non-sexual function for adult–adult play fighting exists among primates is further supported by the strong predictive effect found for social organization. That is, for strepsirrhines, 30–40% of the variance in the distribution of non-sex play was accounted for by the degree of aggregation in social organization: species having more loosely structured social relations are more likely to engage in play (see also Pellis & Iwaniuk 1999b). The findings support the hypothesis that play is an adaptation associated with social relationships in species with infrequent

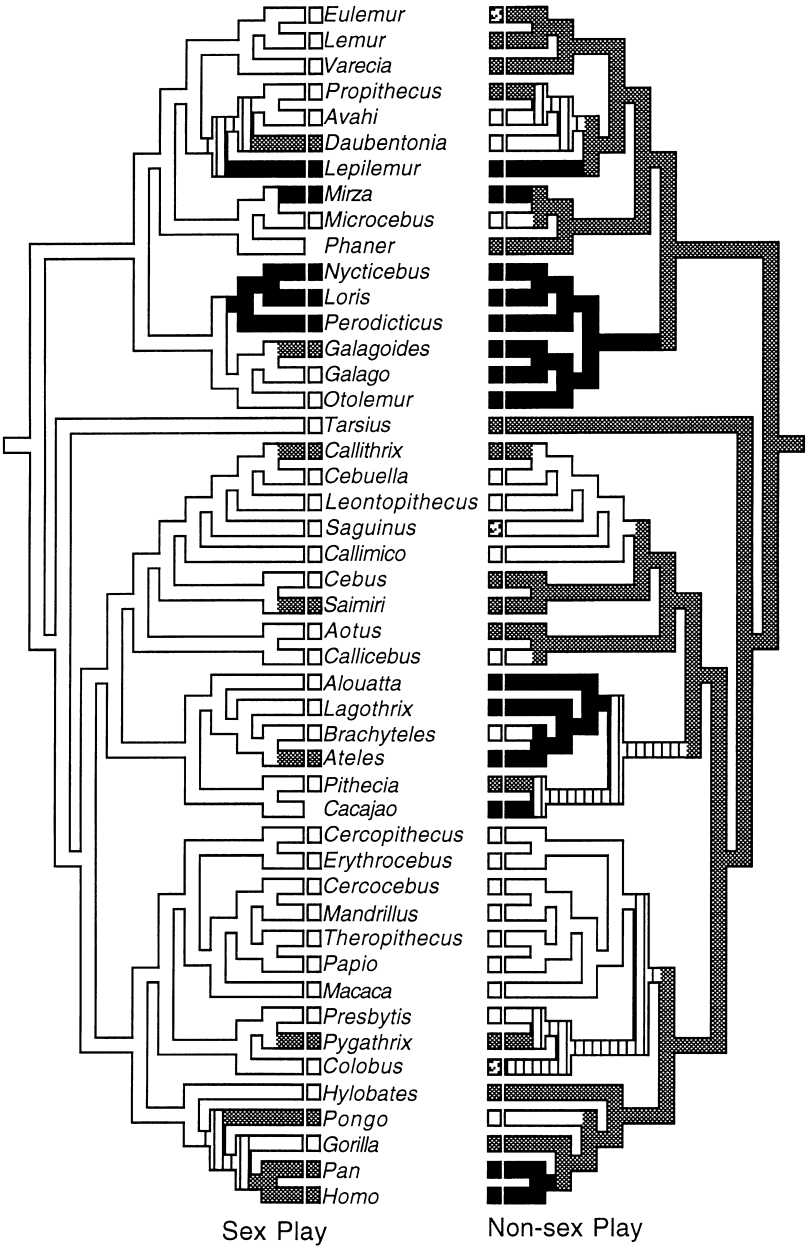


Fig. 2: Cladograms for genera showing the pattern of evolutionary change in sex play (left) and non-sex play (right). Legend: white, rare or absent; speckled, infrequent; grey, moderate; black, common; cross-hatched, equivocal optimization

or erratic social interactions (see below). That the present analysis did not find any evidence that the distribution of adult–adult play is significantly influenced by phylogenetic relationships, adds further support to the likelihood that the distribution of this trait is influenced by socio-ecological factors, and hence has some adaptive value. However, before exploring the implications of our results for the understanding of adult play in primates, some caveats regarding the comparative methods used need to be considered.

There are many reasons for caution in interpreting the meaning of the results obtained by comparative methods (Losos 1999). For example, using different phylogenetic hypotheses can have a significant effect on the results obtained, although such effects may be attenuated. If the changed position of a species alters its branching node, so that it is closer to the base of the tree, then the statistical effect may be greater than if the change is closer to the terminal branches (Purvis & Rambaut 1995; Price 1997).

Another problem is in the coding system used to compare traits among species. There can be a trade-off between consistency and accuracy. For example, in this study, we had to use slightly different criteria to code play, depending on the inclusiveness of the data set. To include more species in a comparison, the rating scale may need to involve less fine-grained distinctions and so involve fewer gradations, thus reducing the variance present. Simpler categories may sometimes lead to ambiguity in comparing species. This is illustrated by *Daubentonia*. From the available literature (Sterling & Richard 1995), it would appear that other than mother–infant association, adults limit their contact to the mating period. So, while it is given a score of 0 as its social organization measure, *Daubentonia* is unlike the other nocturnal strepsirrhines used for this study with such a score. That is, there appears to be little opportunity for adults in this species to come together outside of the mating season. Therefore, a score of 0 for non-sex play in this species is not quite the same as a 0 for *Microcebus murinus*, where some non-sexual association does occur (Martin 1973). A solution to many of these problems is to use multiple methods to assess whether they converge on the same conclusion, thus increasing confidence in those conclusions (Pellis & Iwaniuk 1999a).

A Model for the Evolution of Adult–Adult Play in Primates

Brueggeman (1978) showed that while rare, adult–adult play occurs in rhesus macaques. Based on a detailed evaluation of each context in which such play occurred, Brueggeman (1978) concluded that play was used for social assessment purposes. This conclusion is consistent with a range of studies of various primate and non-primate mammals (Pellis & Pellis 1996). Other than during courtship, play fighting among adolescents, sub-adults and adults appears to be used in two broad contexts. One is in promoting the establishment and maintenance of social bonds (e.g. Newell 1971; Geist 1982; Evans & Poole 1984; Pellis et al. 1993), and the other is in testing opponents and gaining social advantage (e.g. Newell 1971;

Geist 1981; Paquette 1994; Smith et al. 1999). Play fighting during courtship may thus be viewed as one out of a number of possible contexts in which play fighting is used for social assessment (Brueggeman 1978; Pellis & Pellis 1996). A prediction from this proposed function suggests that reduced familiarity between interactants, due to looser social structure and reduced frequency of contact, should significantly increase the occurrence of play. This prediction has been confirmed for play occurring in both sexual (Pellis & Iwaniuk 1999b) and non-sexual contexts (this study).

It would appear that the most likely ancestral state of play for primates was relatively rare to moderate in prevalence. The prevalence of this behavior subsequently increased in some clades and decreased in others (Fig. 2). Even though phylogenetic inertia does not appear to account for the distribution of play in this order, some families appear to be impoverished in this regard (e.g. Cercopithecidae), whereas others appear to be well endowed (e.g. Loridae). This suggests that it is likely that features characteristic of families have either increased or decreased the value of play. Two such features are considered below.

Social organization

Play is least common in Old World monkeys and most common in nocturnal prosimians. Old World monkeys as a group are characterized by highly structured social systems involving long-term, if not life-long, association among group members (Di Fiore & Rendall 1994). In contrast, nocturnal prosimians as a group have the lowest levels of aggregation in their social organization (Charles-Dominique 1995). Therefore, while the social relations among members of a troop of cercopithecids are distinguished by a relatively high degree of predictability, loosely associated lorids are faced with new partners or ones encountered only periodically.

The sizes of social groups are not likely, in themselves, to be sufficient to increase the value of play. For example, both Hamadrayas baboons (*Papio hamadryas*) and uakaris (*Cacajao calvus*) may form troops of 100 or more animals. Hamadryas baboon troops, no matter how large, are composed of highly cohesive subgroups with structured relationships both within and between such subgroups (Kummer 1995). Although little is known of uakari social life, the large troops of this species appear to have a more fluid composition, with many combinations of associations and interactions being possible (Fontaine 1981). Adult–adult play has not been reported as a feature of Hamadryas baboons, whereas such play is highly pronounced in the uakari. Indeed, species such as *Ateles* and *Pan*, with their more fluid social systems (Nishida & Hiraiwa-Hasegawa 1986; Robinson & Janson 1986), are among the anthropoids with the greatest prevalence of play. The use of play appears to be most closely associated with the degree of uncertainty in social relationships, either because of a reduced frequency of encountering conspecifics (e.g. nocturnal prosimians) or because social relationships are relatively fluid (e.g. chimpanzees, spider monkeys).

Patterns of communication

Nocturnal prosimians have a relatively impoverished capacity for moderate to short range communication – with scent marks and calls being most common (Charles-Dominique 1978). Contrast this with a cercopithecid such as the mandrill, *Mandrillus sphinx*, which has a brightly colored face and posterior, a rich repertoire of facial expressions and body postures (Darwin 1872), and a variety of vocalizations – as is the case for cercopithecids in general (Marler 1965). Therefore, there exists for the mandrill a depth and breadth of opportunity for subtle communication in the absence of direct contact. For many nocturnal prosimians, such subtle information about another animal requires some form of tactile assessment (e.g. Epps 1974; Pagès 1978; Goonan 1993).

The diurnal prosimians have secondarily adopted more simian-like patterns of social organization (Martin 1995), and have increased the range of expressive visual signals and bodily ornaments (i.e. badges) in their repertoire of communication signals (Pariente 1979). In terms of short-range visual and auditary signals, primates can be ranked as follows: nocturnal prosimians < diurnal prosimians < New World monkeys < Old World monkeys (e.g. Marler 1965; Moynihan 1976; Pariente 1979; Petter & Charles-Dominique 1979). As shown in Fig. 2, the prevalence of adult–adult play follows the same order, but in reverse. That is, play fighting is most prevalent in some of the nocturnal prosimians. Therefore, a possible factor in the phylogenetic distribution of adult–adult play is the availability of other means of communication which can serve the evaluative and manipulative function that has been suggested above for play.

One form of social contact common to all primates is allogrooming (Sparks 1967). However, among nocturnal prosimians, allogrooming is often intermixed with play, as the grooming by one animal induces grooming in the other. Such mutual grooming can be competitive, as one animal may restrain the other, and the partner may resist such restraint (e.g. Epps 1974; Pagès 1978). In contrast, among simians, allogrooming is a more unidirectional activity, with bouts reciprocated, but not mutually contested (e.g. Barton 1983; Vea et al. 1999). Furthermore, there are gradations in the frequency of occurrence of allogrooming: it is less common in many prosiminans, New World monkeys and apes, and most common in Old World monkeys (Sparks 1967). This also suggests a reciprocal relationship with the occurrence of play. While quantitative comparisons are needed for the evaluation of play with regard to allogrooming and close-quarter communicatory signals, this qualitative review suggests that play may be most valuable when the signals are few and the allogrooming is less frequent and structured.

The Function of Play

Even though rare in cercopithecids, some species appear to use play in adult–adult interactions extensively (Rhine 1973). Also, in species for which such play is rare, when it does occur, it is used in the kinds of social assessment contexts described above (Brueggeman 1978). Finally, in species generally not known to exhibit adult–adult play, the adults may use play fighting in their social interac-

tions with non-adults, which in turn may influence their social relations with other adults. For example, among olive baboons (*Papio anubis*), adult males seem to form special friendships with specific females and, even when they are not the father, spend time affiliating with the females' offspring. Such affiliative contact includes allogrooming and play (Smuts 1985). So while the play option for social interaction may have been reduced in importance for cercopithecids, its use has not been completely eliminated. Furthermore, the context of its use underscores its function as a means of social assessment and manipulation.

Given the disproportionately high frequency of studies on cercopithecids (Di Fiore & Rendall 1994; Strier 1994), the view of play as a mainly juvenile phenomenon may not be typical of primates in general. For example, in a captive troop of spider monkeys (*Ateles geoffroyi*), over 30% of the play recorded involved adults playing with other adults (Pellis & Pellis 1997). In an even more extreme case, a study of captive slow loris (*Nycticebus coucang*) showed that play was more frequent among adults than juveniles (Ehrlich & Musicant 1975). Therefore, it would appear that the cercopithecids are the most derived family with regard to adult–adult play, as they appear to be for most other behavioral and morphological traits (Fleaggle 1988). Even so, in terms of the contexts of its use, the remnant occurrence of such play in this family is consistent with the pattern seen in other primates.

Among adolescents and sub-adults, play fighting appears to be more commonly reported as being used for evaluative and social manipulative purposes, both in primates (e.g. Neill 1976; Paquette 1994; Reichard & Sommer 1997) and other mammals (e.g. Geist 1981, 1982; Mills 1990; Pellis et al. 1993). This raises the possibility that, during the juvenile phase, play fighting may provide the opportunity to refine these evaluative and manipulatory skills (Biben 1998; Pellis & Pellis 1998a). For groups such as cercopithecids, where the occurrence of adult–adult play is greatly reduced, the play of juveniles could be used to test this hypothesis against competing hypotheses. A commonly espoused functional hypothesis for juvenile play fighting is that it is used for refining combat skills (e.g. Aldis 1975; Symons 1978; Fagen 1981; Smith 1982). If this is the case, then the organization of play fighting should primarily emphasize the species-typical tactics of attack and defense. However, if play fighting in juvenile cercopithecids is a remnant of an ancestral adult behavior used for social assessment, then the organization of the play should reflect species differences in social organization and patterns of communication.

Conclusions

The data are consistent with the possibility that adult–adult play first arose due to its functional value in non-sexual social contexts, but once present, its use was co-opted for courtship. The observations, that both courtship play (Pellis & Iwaniuk 1999b) and play fighting in general (this study) are more likely to occur in social contexts of low familiarity between the interactants, support the hypothesis that play fighting is used for social evaluative and manipulative purposes (Brueg-

geman 1978). Small, nocturnal and relatively non-gregarious ancestral primate species, with a relatively impoverished repertoire of visual and auditory signals (Charles-Dominique 1995; Martin 1995), would have had play fighting as a close-quarter means of gaining information and manipulating conspecifics through tactile and olfactory modalities. Divergence from the ancestral traits may have led to a reduction in the value of play for such communication purposes. Future research should focus on defining more precisely the contexts in which play may best serve these functions (Pellis & Pellis 1996) and on identifying the mechanisms by which play achieves these goals (Biben 1998; Pellis & Pellis 1998a). Finally, the structure of play in juveniles needs to be re-evaluated in light of the functional properties of adult play.

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